The Thirsty Cloud

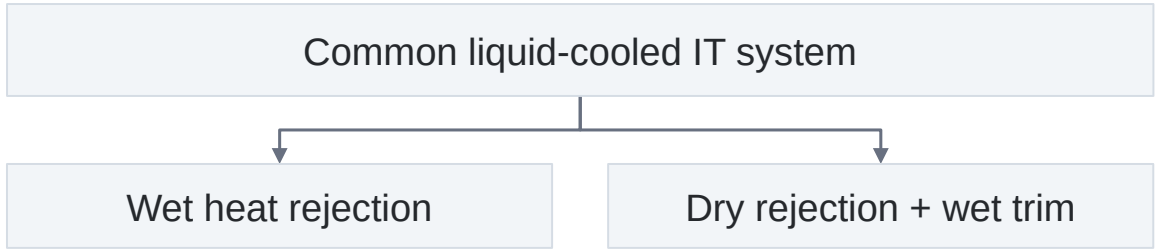
Water supply and cooling choices for Singapore's data-centre growth

DSS5202 Group 29 (2026) Aditya Vijindra Pandhari Apurwa Kumari Vanshika Dhamija

1. Decision and comparison basis

We compare additional water supply with lower water use for new data centres during 2027–2035. All options serve 1 MWh of IT energy and share the same liquid-cooled IT baseline.

Sources: PUB (2025), EMA (2026) and team assumptions.



A1 Desalination
A2 NEWater
A3 NEWater + tower reuse

A4 NEWater for remaining demand

2. Additional water demand in 2035

2.58 million m³/year: 200 MW case
9.01 million m³/year: 700 MW hypothetical case

Wet rejection at 70% utilisation after the assumed ramp.
Water reduction vs A2: A3 30%, A4 76%.

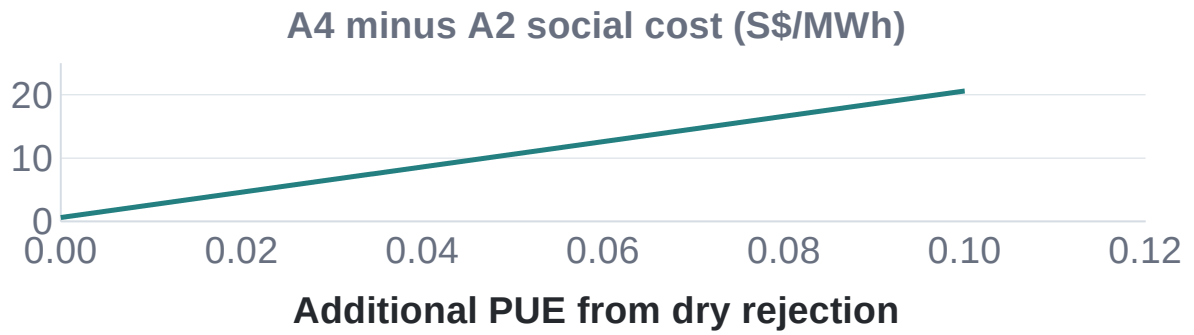
Methods: LCA, LCC and AHP with sensitivity tests.
Balanced AHP ranks A3 first with illustrative weights.

3. Base-case results per MWh of IT

Metric	A1	A2	A3	A4
Water (m³)	2.10	2.10	1.46	0.50
GHG (kg CO_2e)	3.42	0.99	0.74	19.71
Social cost (S$)	2.60	1.00	1.54	11.60
Operator cost (S$)	6.80	5.25	4.83	15.64

Partial boundary: water supply and incremental heat rejection. Costs exclude carbon monetisation.

4. Cost of reducing water use



A2 minimises base social cost. A4 minimises water use. Verify dry-rejection electricity at the site.